Hidden Region Finder

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Abstract

Hidden regions are contributions to asymptotic expansions that are missed by the ordinary Newton-polytope construction because nominally superleading monomials cancel on a positive Landau pinch. We introduce the Hidden Region Finder, a parameter-space algorithm which identifies the cancellation ideal, separates the superleading polynomial from its obstruction, determines the region scaling and certifies the resolved leading Lee–Pomeransky polynomial. The construction includes polynomial cancellation factors, contraction-boundary regions, asymptotic-order alignment and successive cancellation layers. Applications to massless four-, five- and six-point integrals reveal a sparse topological organisation. Through four loops, every hidden region in the audited wide-angle and three-channel Regge samples is associated with the three-loop Crown, in the interior or as a contraction minor, while sixteen mixed-sign topologies with no Crown minor have none. The SuperCrown and HyperCrown exhibit distinct codimension-one and codimension-two descendants of this seed. At five and six points, the same special topology can support different hidden regions in different kinematic expansions. Thus the algorithm provides both a region finder and a tool for exposing relations between scattering limits.

Contents

1 Introduction

The Method of Regions (MoR) constructs the asymptotic expansion of a multiscale Feynman integral as a sum of integrals associated with distinct scalings of the integration variables. Its geometric formulation in parameter space maps every monomial of the Lee–Pomeransky polynomial to an exponent point. Lower facets of the resulting Newton polytope determine the ordinary endpoint regions and their scaling vectors.

The Newton polytope retains the monomial exponents but not the information carried by their coefficients. In a physical Minkowski region, monomials of opposite sign can cancel at positive values of the parameters. If the cancellation is reinforced by a Landau pinch, a set of monomials may be individually more singular than the physical leading contribution while their sum is suppressed. The corresponding scaling is not a lower-facet normal of the original polynomial and is therefore hidden from the ordinary geometric construction. We call it a *hidden region* (HR).

A hidden region is specified by more than a scaling vector. One must identify the common positive cancellation locus, determine which monomials form the superleading sector, resolve their cancellation against the remaining Lee–Pomeransky polynomial, and verify that the resulting leading integral is scaleful. The Hidden Region Finder (HRF) is the systematic parameter-space construction that performs these tasks.

The present paper concerns massless internal propagators only. This is a structural restriction, not a choice of examples: both Symanzik polynomials are then multiaffine, or equivalently linear in every individual edge parameter. Internal masses add $\mathcal{U}\sum_e m_e^2 x_e$ to $\mathcal{F}$, which can generate quadratic dependence on an individual x_e. Hidden regions in that more general setting are important, but require a separate extension of the construction and will not be treated here.

The presentation has two parts. Part I defines the construction without reference to a particular graph. Part II develops the same ideas through a sequence of massless examples. Each example introduces one additional phenomenon: a factorised cancellation with uniform scaling; an obstruction that fixes a non-uniform scaling; non-binomial cancellation factors; and finally composite limits for which preliminary asymptotic-order alignment and, in some cases, more than one cancellation layer are required. The examples also reveal a recurring physical organisation: a special seed topology can support different HRs in different kinematic expansions.

Part I
Parameter-space formulation

2 Parametric representation and notation

Consider a graph $\mathcal{G}$ with N massless internal propagators and Lee–Pomeransky (LP) polynomial

$$\mathcal{P}(\boldsymbol{x}, \boldsymbol{s}) = \mathcal{U}(\boldsymbol{x}) + \mathcal{F}(\boldsymbol{x}; \boldsymbol{s}). \tag{1}$$

The spanning-tree and spanning-2-forest formulae make the relevant structure immediate. For an L-loop graph, $\mathcal{U}$ is homogeneous of degree L, while $\mathcal{F}$ is homogeneous of degree $L+1$ in the edge parameters. Each edge occurs at most once in every complementary tree or 2-forest product. Hence, for massless internal propagators, both graph polynomials are linear in each individual x_e. This multiaffinity is a standing assumption below. Up to a normalisation which is irrelevant here, its integral has the form

$$I_{\mathcal{G}} = \mathcal{N}_{\mathrm{LP}} \int_0^\infty \prod_{e=1}^{N} \frac{\mathrm{d}x_e}{x_e}\, x_e^{\nu_e}\, \mathcal{P}(\boldsymbol{x}, \boldsymbol{s})^{-D/2}. \tag{2}$$

The N variables x_e are independent Lee–Pomeransky edge parameters, each integrated over $(0, \infty)$. In particular, there is no constraint on $\sum_e x_e$, or on the sum of any subset of the parameters. We use x_e rather than the conventional Feynman-parameter symbol α_e.

An optional, distinct projective representation can be obtained by introducing one additional parameter x_0 and the homogeneous polynomial

$$\widehat{\mathcal{P}}(x_0, \boldsymbol{x}; \boldsymbol{s}) = \mathcal{F}(\boldsymbol{x}; \boldsymbol{s}) + x_0 \mathcal{U}(\boldsymbol{x}). \tag{3}$$

There are then $N+1$ parameters, and their common rescaling may be fixed by inserting $\delta(1 - E(x_0, \boldsymbol{x}))$ for any positive degree-one homogeneous function E. This homogenised embedding is distinct from the non-projective LP integration space in (2) and is not assumed here.

Let $\delta \to 0^+$ define the kinematic expansion, and let $\boldsymbol{s}$ denote a set of real kinematic coordinates. No sign convention is imposed on the individual components of $\boldsymbol{s}$; a physical domain $\mathcal{K}$ is specified by the appropriate inequalities among them. This allows the same variables to be retained across different physical regions, making changes of sign under crossing or analytic continuation explicit. A separate positive parametrisation of a particular domain may be introduced locally as a bookkeeping aid for proving inequalities or performing semialgebraic sign tests, but neither the Newton-polytope construction nor the first-sheet pinch condition requires it. Retaining signed common variables also makes a change in the existence of a cancellation directly visible in a factorised superleading polynomial. After expanding the kinematics, write

$$\mathcal{P}(\boldsymbol{x}, \delta; \boldsymbol{s}) = \sum_i m_i, \qquad m_i \equiv c_i(\boldsymbol{s})\, \delta^{a_i} \boldsymbol{x}^{\boldsymbol{r}_i}, \qquad \boldsymbol{x}^{\boldsymbol{r}_i} = \prod_{e=1}^{N} x_e^{r_{i,e}}. \tag{4}$$

Thus $\boldsymbol{r}_i = (r_{i,1}, \ldots, r_{i,N})$ is the length-N LP-parameter exponent vector of the monomial m_i, while its augmented length-$N+1$ exponent point is

$$\vec{r}_i = (\boldsymbol{r}_i; a_i) \in \mathbb{R}^{N+1}. \tag{5}$$

No fixed sign of m_i on the whole physical domain is assumed. Although $\delta^{a_i} \boldsymbol{x}^{\boldsymbol{r}_i} > 0$ for $\delta > 0$ and $\boldsymbol{x} > 0$, the coefficient $c_i(\boldsymbol{s})$ may vanish and change sign within or between the domains under comparison. The same qualification applies if several literal monomials have been grouped into a kinematic coefficient or polynomial term. A region vector is represented by an augmented normal whose last component is normalised to one,

$$\vec{v}_R = (\boldsymbol{v}_R; 1), \qquad x_e \longmapsto \delta^{v_{R,e}} x_e, \tag{6}$$

and the weight of a monomial is

$$w_R(m_i) = \vec{v}_R \cdot \vec{r}_i = a_i + \boldsymbol{v}_R \cdot \boldsymbol{r}_i. \tag{7}$$

The last component is always displayed when a vector is meant to be a normal in augmented exponent space. A list containing only the first N entries is an LP-parameter scaling and not a fully normalised augmented normal.

For a cancellation-free polynomial, the monomials of minimum weight form a lower face of its Newton polytope. A codimension-one lower face has an inward-pointing normal whose last component is positive; after normalising that component to one, the normal gives the corresponding facet-region scaling.

3 Hidden regions and positive pinches

The LP variables are integrated on the positive contour $\mathcal{C}_{\mathrm{LP}} = \mathbb{R}^N_{>0}$. An interior first-sheet pinch of a polynomial $\mathcal{Q}(\boldsymbol{x}; \boldsymbol{s})$ is therefore a solution

$$\mathcal{Q} = 0, \qquad \frac{\partial \mathcal{Q}}{\partial x_e} = 0 \quad (e = 1, \ldots, N), \qquad \boldsymbol{x} \in \mathbb{R}^N_{>0}, \quad \boldsymbol{s} \in \mathcal{K}. \tag{8}$$

For a homogeneous polynomial, the derivative equations imply $\mathcal{Q} = 0$ by Euler's identity. On a contraction stratum the same equations are imposed only on the active, nonzero parameters.

A cancellation in the positive orthant requires terms of both signs. Consequently, every active derivative at an interior pinch must either vanish identically or contain terms of both signs. This elementary necessary condition is useful as a rapid prefilter, but it neither identifies the common pinch locus nor proves that a hidden region exists. The defining conditions below first use the scaling law to isolate $\mathcal{F}_{\mathrm{SL}}$, and then require a simultaneous positive solution of its derivative equations.

4 Definition of a hidden region

A HR is characterised by a superleading monomial support $\mathcal{S}_{\mathrm{SL}}$, its associated polynomial

$$\mathcal{F}_{\mathrm{SL}} = \sum_{i \in \mathcal{S}_{\mathrm{SL}}} m_i, \tag{9}$$

and a final scaling

$$\vec{v}_{\mathrm{HR}} = (\boldsymbol{v}_{\mathrm{HR}}; 1) \tag{10}$$

such that:

(a) every monomial indexed by $i \in \mathcal{S}_{\mathrm{SL}}$ has the same individual weight,

$$w_{\mathrm{HR}}(m_i) = W_{\mathrm{SL}}, \qquad i \in \mathcal{S}_{\mathrm{SL}}; \tag{11}$$

(b) $\mathcal{F}_{\mathrm{SL}}$ has a common first-sheet pinch in $\mathcal{K}$,

$$\left.\frac{\partial \mathcal{F}_{\mathrm{SL}}}{\partial x_e}\right|_{\boldsymbol{x}>0,\, \boldsymbol{s}\in\mathcal{K}} = 0 \qquad \text{for every active } x_e; \tag{12}$$

(c) the cancellation suppresses the sum of the superleading monomials to the first nonvanishing LP layer of weight W_{HR}, with

$$W_{\mathrm{HR}} > W_{\mathrm{SL}}; \tag{13}$$

(d) after resolving the cancellation, the leading LP polynomial defines a scaleful lower-facet region.

The gap $W_{\mathrm{HR}} - W_{\mathrm{SL}}$ measures the depth of the cancellation. It is a physical output of the full problem and is not a freely chosen normalisation.

The superleading sector is not defined merely by mixed signs. Mixed signs are necessary for an interior pinch, but the simultaneous positive solution, the hierarchy, and the final lower-facet condition are all essential.

Principle 1 (Scaling precedes the positive pinch test)**.** *The mixed-sign test is applied to derivatives of the starting polynomial $\mathcal{F}_\star$, but the simultaneous Landau equations are not. The weight homogeneity and separation conditions must first isolate the support $\mathcal{S}_{\mathrm{SL}}$. The positive first-sheet equations are then imposed on the resulting polynomial $\mathcal{F}_{\mathrm{SL}}$. Generically $\partial\mathcal{F}_\star/\partial x_e = 0$ has no simultaneous positive solution because $\mathcal{F}_\star$ still contains the obstruction and other terms excluded by the HR scaling.*

5 Cancellation ideals and the obstruction

Before any LP-parameter scaling, expand

$$\mathcal{F}(\boldsymbol{x}, \delta; \boldsymbol{s}) = \mathcal{F}_0(\boldsymbol{x}; \boldsymbol{s}) + \sum_{n>0} \delta^n \mathcal{F}_n(\boldsymbol{x}; \boldsymbol{s}). \tag{14}$$

When the relevant superleading monomials are contained in $\mathcal{F}_0$, the cancellation analysis begins with $\mathcal{F}_\star = \mathcal{F}_0$.

5.1 Cancellation factors and their common locus

The derivatives of $\mathcal{F}_\star$ are inspected for primitive, non-monomial polynomial factors $f_j(\boldsymbol{x}, \boldsymbol{s})$ which can vanish for $\boldsymbol{x} > 0$ and $\boldsymbol{s} \in \mathcal{K}$. At fixed $\boldsymbol{s} \in \mathcal{K}$, a necessary condition for such a zero is that the nonzero coefficients of the LP-parameter monomials are not all of the same sign. A monomial factor such as x_e is not a cancellation hypersurface in the interior and is removed from the primitive factor.

This inspection generates candidate cancellation structures; it is not a Landau test on $\mathcal{F}_\star$. In particular, the full derivative system of $\mathcal{F}_\star$ is not required to have a positive solution at this stage and generically does not. A proposed relative scaling must first select a homogeneous superleading support $\mathcal{S}_{\mathrm{SL}}$ and strictly separate it from the obstruction and all other monomials.

Only for the selected polynomial $\mathcal{F}_{\mathrm{SL}}$ is the simultaneous first-sheet pinch condition imposed. When its derivative equations reduce to a set of primitive cancellation factors, physical admissibility requires a common positive zero at one and the same kinematic point:

$$\exists\,(\boldsymbol{x}, \boldsymbol{s}) \in \mathbb{R}^N_{>0} \times \mathcal{K} : \qquad f_1(\boldsymbol{x}, \boldsymbol{s}) = \cdots = f_r(\boldsymbol{x}, \boldsymbol{s}) = 0. \tag{15}$$

Equivalently one solves (12) directly. The common locus, rather than any one factor in isolation, encodes the pinch. The elementary mixed-sign condition can discard impossible ingredients, but it cannot replace the scaling selection of $\mathcal{S}_{\mathrm{SL}}$ or the subsequent simultaneous positive solution.

5.2 Generators and the obstruction decomposition

Candidate generators are products of genuine cancellation polynomials,

$$g_k = \prod_{f_j \in S_k} f_j, \tag{16}$$

with at least two cancellation factors in the generator structures used by HRF. The factors, not monomial edge parameters, define the singular hypersurfaces. For a generator set $G = \{g_1, \ldots, g_r\}$, define

$$\mathcal{I}_{\mathrm{canc}}(G) = \langle g_1, \ldots, g_r \rangle. \tag{17}$$

The HRF seeks a decomposition

$$\mathcal{F}_\star = \mathcal{F}_{\mathrm{SL}} + \mathcal{F}_{\mathrm{Obs}}, \qquad \mathcal{F}_{\mathrm{SL}} \in \mathcal{I}_{\mathrm{canc}}(G), \tag{18}$$

or equivalently

$$\mathcal{F}_{\mathrm{SL}} = \sum_k M_k(\boldsymbol{x}, \boldsymbol{s}) g_k. \tag{19}$$

The polynomial $\mathcal{F}_{\mathrm{Obs}}$ is the obstruction: it prevents the full starting polynomial from vanishing on the candidate cancellation locus but is to be suppressed relative to the individual SL monomials by the HR scaling.

Different generator presentations can describe the same ideal and the same pinch locus. The presentation is a discovery device and does not label a physical region. Polynomial multipliers M_k in (19) are allowed; a special factorised presentation valid for one topology must not be promoted to a general rejection rule.

6 Asymptotic-order alignment: selecting the correct starting face

6.1 Why $\mathcal{F}_0$ need not be the correct starting point

In a composite limit, the relevant cancellation structure may not be visible in the naive fixed-$\boldsymbol{x}$ leading polynomial $\mathcal{F}_0$. A preliminary LP-parameter rescaling can expose a different face of the full δ-dependent polynomial before the cancellation decomposition is applied. This operation is called *asymptotic-order alignment*.

The rescaling changes the effective expansion order of a monomial from a_i to $a_i + \boldsymbol{\phi} \cdot \boldsymbol{r}_i$. Monomials that belong to different fixed-$\boldsymbol{x}$ orders in δ can thereby acquire a common effective weight and enter the same cancellation sector. This cross-order alignment, rather than the mere selection of a different face, is the purpose of the operation.

Let $\boldsymbol{\phi} \in \mathbb{Q}^N$ be the preliminary alignment vector and introduce new parameters $\boldsymbol{y}$ by

$$x_e = \delta^{\phi_e} y_e. \tag{20}$$

For a monomial $m_i = c_i(\boldsymbol{s})\delta^{a_i}\boldsymbol{x}^{\boldsymbol{r}_i}$, define

$$\omega_\phi(m_i) = a_i + \boldsymbol{\phi} \cdot \boldsymbol{r}_i. \tag{21}$$

The order-aligned starting polynomial is the exposed face

$$\mathcal{F}^{[\phi]} = \sum_{\substack{m_i \in \mathcal{F} \\ \omega_\phi(m_i) = \min_{m_j \in \mathcal{F}} \omega_\phi(m_j)}} m_i. \tag{22}$$

The general HRF starting object is therefore

$$\mathcal{F}_\star = \begin{cases} \mathcal{F}_0, & \text{ordinary search,} \\ \mathcal{F}^{[\phi]}, & \text{order-aligned search.} \end{cases} \tag{23}$$

The same cancellation-factor, positive-pinch and obstruction analysis is then applied to $\mathcal{F}_\star$.

Asymptotic-order alignment is not itself the hidden-region scaling. It is the first of two coordinate transformations. It exposes the polynomial on which monomials from different original δ-orders can participate in one cancellation; HRF then determines the scaling about that cancellation locus in the aligned coordinates.

6.2 The second vector and the composition law

Let $\boldsymbol{\rho} \in \mathbb{Q}^N$ denote the HRF scaling in the aligned variables,

$$y_e = \delta^{\rho_e} z_e. \tag{24}$$

Combining Eqs. (20) and (24) gives

$$x_e = \delta^{\phi_e + \rho_e} z_e, \qquad \boldsymbol{v}_{\text{HR}} = \boldsymbol{\phi} + \boldsymbol{\rho}, \qquad \vec{v}_{\text{HR}} = (\boldsymbol{\phi} + \boldsymbol{\rho}; 1). \tag{25}$$

Only the N edge-parameter exponents are added. The last component is the normalisation of the single expansion parameter and is appended once after the two transformations have been composed. Thus one must never add $(\boldsymbol{\phi}; 1)$ and $(\boldsymbol{\rho}; 1)$, which would spuriously produce a last component equal to two.

The composition can be checked without reference to any factorisation. The same original monomial has the successive weights

$$a_i + \boldsymbol{\phi} \cdot \boldsymbol{r}_i, \qquad a_i + (\boldsymbol{\phi} + \boldsymbol{\rho}) \cdot \boldsymbol{r}_i. \tag{26}$$

The second expression, evaluated for every monomial of the complete LP polynomial, is the absolute weight relevant to the physical region. Stage-local weights obtained after an overall power of δ has been factored from $\mathcal{F}^{[\phi]}$ cannot be added to, or directly identified with, these absolute weights.

There is one important qualification. A run on the isolated aligned face may first return a provisional vector $\boldsymbol{\rho}^{(0)}$. This is the correct second vector only if restoration of the complete LP polynomial does not promote further occupied layers. In the general case HRF must restore all layers, compute their orders in the cancellation ideal, and solve the resolved hierarchy again. The resulting $\boldsymbol{\rho}$, rather than $\boldsymbol{\rho}^{(0)}$, is the vector to use in Eq. (25). Successive cancellation layers can change a non-uniform component of the second vector; their effect is not, in general, reducible to an overall shift.

6.3 Uniform-shift ambiguity

The second Symanzik polynomial is homogeneous of fixed parameter degree. Consequently,

$$\boldsymbol{\phi} \longmapsto \boldsymbol{\phi} + c(1, \ldots, 1) \tag{27}$$

adds the same amount to the weight of every monomial of $\mathcal{F}$ and does not change the face $\mathcal{F}^{[\phi]}$. The preliminary alignment vector is therefore an equivalence class, not a unique vector.

The coordinate transformation itself shows how this ambiguity is removed. For fixed original $\boldsymbol{x}$, replacing $\boldsymbol{\phi}$ by $\boldsymbol{\phi} + c\mathbf{1}$ replaces the aligned coordinates by $\boldsymbol{y}' = \delta^{-c}\boldsymbol{y}$. The same second transformation is therefore represented by

$$\boldsymbol{\rho}' = \boldsymbol{\rho} - c\mathbf{1}, \qquad (\boldsymbol{\phi} + c\mathbf{1}) + \boldsymbol{\rho}' = \boldsymbol{\phi} + \boldsymbol{\rho}. \tag{28}$$

Hence the physical total vector is independent of the chosen alignment representative, provided the second vector is transformed consistently.

In practice this compensation is fixed by the complete LP polynomial. The polynomials $\mathcal{U}$ and $\mathcal{F}$ have degrees L and $L+1$, respectively, so a uniform shift changes their relative weights. Requiring the resolved cancellation layer to balance the appropriate $\mathcal{U}$ layer, while satisfying the hierarchy conditions below, fixes the representative and the physical gap $W_{\mathrm{HR}} - W_{\mathrm{SL}}$. This gap is a property of the region and is not an adjustable normalisation.

The consistent two-stage procedure is therefore:

(i) choose any convenient representative $\boldsymbol{\phi}$ that exposes the required face;

(ii) run HRF in the aligned variables to obtain the cancellation ideal and a provisional $\boldsymbol{\rho}^{(0)}$;

(iii) restore $\mathcal{U}$ and every occupied layer of the original δ-dependent $\mathcal{F}$, and determine the corrected $\boldsymbol{\rho}$ from the ideal-jet hierarchy;

(iv) compose the edge exponents according to $\boldsymbol{v}_{\mathrm{HR}} = \boldsymbol{\phi} + \boldsymbol{\rho}$, append the final component one, and verify the resolved lower facet and scalefulness.

7 Successive cancellation layers and the final vector

7.1 The complete LP polynomial is decisive

For a candidate final vector, organise the complete polynomial into its actually occupied weight layers,

$$\mathcal{P}(\delta^{\boldsymbol{v}}\boldsymbol{x}, \delta; \boldsymbol{s}) = \sum_{q\in\Lambda} \delta^q \, \mathcal{P}_q(\boldsymbol{x}, \boldsymbol{s}), \tag{29}$$

where Λ is the set of weights which occur. One must not insert fictitious intermediate layers. For example, if a parametrisation produces only even powers of the expansion parameter, the absence of an odd power is not an additional cancellation.

Let

$$\mathcal{I} = \langle f_1, \ldots, f_r \rangle \tag{30}$$

be the ideal generated by independent cancellation factors defining the common pinch. A layer $\mathcal{P}_q$ which lies in $\mathcal{I}$ vanishes on the pinch; a layer in $\mathcal{I}^m$ vanishes to at least m-th order in transverse directions. The relevant information is its leading nonzero class in the filtration

$$\mathcal{I}^m/\mathcal{I}^{m+1}. \tag{31}$$

Locally, take transverse coordinates $y_j = f_j$. If the approach to the common pinch scales as

$$y_j \sim \delta^{\tau_j}, \qquad \tau_j > 0, \tag{32}$$

then a jet $y_1^{\beta_1} \cdots y_r^{\beta_r}$ in the q-th occupied layer has effective weight

$$q + \sum_j \beta_j \tau_j. \tag{33}$$

The first resolved nonvanishing contribution, after accounting for these transverse weights and the $\mathcal{U}$ balance, defines W_{HR}. It need not be the first nominal obstruction layer. A second or later layer may also vanish on the same singular locus.

All independent transverse coordinates must be suppressed with positive weights. A solution which suppresses only a proper subset of the cancellation factors does not certify the common pinch under consideration. Likewise, an unresolved continuous family signals that the total scaling has not yet been fixed and may correspond to a residual scaleless rescaling.

7.2 Hierarchy conditions

The final, normalised vector $\vec{v}_{\mathrm{HR}} = (\boldsymbol{v}_{\mathrm{HR}}; 1)$ must first satisfy

$$w_{\mathrm{HR}}(m_i) = W_{\mathrm{SL}}, \qquad m_i \in \mathcal{F}_{\mathrm{SL}}. \tag{34}$$

For the general, layered problem, the remaining hierarchy is a condition on the resolved jets rather than on the nominal weights of individual monomials. If $y^{\boldsymbol{\beta}} h_{\boldsymbol{\beta}}$ is a nonzero leading jet of an occupied layer $\mathcal{P}_q$, then

$$q + \boldsymbol{\beta} \cdot \boldsymbol{\tau} \geq W_{\mathrm{HR}} > W_{\mathrm{SL}}, \tag{35}$$

and equality must occur for at least one resolved contribution. A nominal layer with $q < W_{\mathrm{HR}}$ is therefore allowed only if its positive ideal order supplies enough transverse suppression to satisfy (35). This is precisely how a second cancellation layer can occur.

If no intermediate ideal-valued layer is present, every term outside $\mathcal{F}_{\mathrm{SL}}$ has zero transverse ideal order. In that special case (35) reduces to the single-layer monomial condition

$$w_{\mathrm{HR}}(m_i) \geq W_{\mathrm{HR}} > W_{\mathrm{SL}}, \qquad m_i \in \mathcal{P} \setminus \mathcal{F}_{\mathrm{SL}}. \tag{36}$$

Equations (34)–(35), the positive transverse weights, and the balance with $\mathcal{U}$ determine the physical cancellation depth and remove the uniform alignment ambiguity.

Terms suppressed at fixed LP parameters must be restored before this test. If a term from $\mathcal{F}_n$, $n > 0$, is promoted to the SL level or below by the candidate scaling, then the proposed $\mathcal{F}_{\mathrm{SL}}$ is incomplete and the analysis must be repeated with that term included.

8 Scalefulness and dissection

Whether the region integral is scaleful is a property of the resolved monomial support of the LP polynomial. It does not depend on the space-time dimension D. A mere count of variables appearing in selected monomials is a useful diagnostic but is not a general certificate.

After the cancellations are resolved, the decisive geometric requirement is that the leading support define a lower facet with an inward-pointing normal whose last component is positive. There are two cases.

(1) If the monomials at the resolved $W_{\rm HR}$ layer already exhibit the required lower facet in the original coordinates, this is a sufficient certificate and no change of variables is needed.

(2) If that test is inconclusive, or if the full-layer equations do not fix a unique total vector, one must use local coordinates adapted to the pinch. A dissection separates the integration domain into sectors and maps the cancellation hypersurfaces to endpoints. In each sector the polynomial is cancellation-free, so the ordinary Newton-polytope facet criterion is both the safe region test and the direct route to the region integral.

The facet normal obtained after dissection is pulled back to the original LP coordinates and then normalised with last component one. This pullback is the authoritative total vector. It may differ from the raw alignment-plus-relative-HRF sum because the latter retains the ambiguity described in (27) and may not have resolved all cancellation layers.

Dissection is therefore not required for every discovery. A unique ideal-layer solution supplies the same certification in the original coordinates. Dissection remains the necessary fallback when the local ideal geometry is nonlinear, singular or non-unique, and it is often useful later for evaluating the HR integral.

9 From parameter scaling to momentum flow

A parameter-space vector fixes propagator virtualities, but it does not by itself fix a unique routing or the separate light-cone components of every edge momentum. We therefore distinguish an *edge-flow assignment* from a *mode-complete momentum region*. The former records the powers of all propagator momenta at a candidate pinch. The latter additionally specifies a basis of independent loop fluctuations and the widths of their integration domains.

Choose a light-cone frame and write

$$q_e \sim (\delta^{a_e}, \delta^{b_e}, \delta^{c_e}) \equiv (q_e^+, q_e^-, |q_{e\perp}|). \tag{37}$$

If the certified LP vector is $\boldsymbol{v}$, inverse Schwinger scaling gives the relative virtuality powers

$$q_e^2 \sim \delta^{\kappa_e}, \qquad \kappa_e = \kappa_{\rm ref} - v_e, \tag{38}$$

where one known physical virtuality fixes $\kappa_{\rm ref}$. For an ordinary direct realization,

$$\kappa_e = \min(a_e + b_e, 2c_e). \tag{39}$$

When $a_e + b_e = 2c_e < \kappa_e$, the longitudinal and transverse terms in $q_e^2 = q_e^+ q_e^- - q_{e\perp}^2$ must cancel at leading order. Such a near-on-shell enhancement requires a coefficient-level check and cannot be certified by valuations alone.

At every vertex and for every component, momentum conservation requires the smallest exponent in the signed sum to occur at least twice. This tropical condition is necessary, but by itself is not sufficient: it does not test the signs and leading coefficients. Our reconstruction therefore proceeds as follows.

9.1 Ordinary facet regions

For a facet region the working assumptions, supported by the standard soft and collinear examples, are:

(i) use the facet vector and an exact external-momentum chart to determine the target κ_e and external component powers;

(ii) solve Eq. (39) together with tropical vertex conservation for every edge;

(iii) prefer branches in which the largest possible number of propagators realize their virtuality directly and homogeneously, $a_e + b_e = 2c_e = \kappa_e$;

(iv) choose independent loop variables as chords of a spanning tree, or an equivalent affine rerouting by external momenta, and verify that their component scalings reproduce the complete edge table;

(v) power count with the fluctuation widths of these independent loop variables, not with a selected subset of edge momenta.

For the facet examples considered below, generic leading coefficients then realize the valuation solution and no additional cancellation-domain support is required.

9.2 Hidden regions

For an HR, the same steps are necessary but not sufficient. The emerging picture is:

(i) use the certified *total* vector, including asymptotic-order alignment, to determine the virtuality powers;

(ii) construct a complete edge-flow table satisfying virtualities and vertex conservation;

(iii) impose the cancellation ideal, or equivalently local coordinates transverse to the positive pinch, at coefficient level. This selects the correlated branch and fixes the restricted support that valuations alone cannot see;

(iv) search all independent loop bases and affine reroutings. A loop mode is Glauber only when an independent integration variable has $a + b > 2c$, so that its transverse virtuality dominates. A transverse-dominated *value* of a combination is evidence for a Glauber direction, but is not yet its fluctuation measure;

(v) for each potentially restricted light-cone component, express every Feynman denominator in the adapted loop basis and retain those whose dependence on the local fluctuation is leading. Their $i0$ prescriptions determine whether poles approach from opposite half-planes and fix the fluctuation width;

(vi) determine the remaining independent loop-component widths in the local pinch coordinates and compare the resulting momentum-space power count with the parameter-space count.

The last comparison is a strong certificate because the cancellation depth appears as restricted integration support in both representations. We shall call a result *mode complete* only after this width information is known. A tropical edge-flow table without the coefficient and width checks is a necessary momentum-space candidate, not a proof of the region.

Concretely, suppose a propagator is linear in the selected component,

$$D_e = A_e \ell^+ + B_e + i0, \qquad \ell_e^+ = -\frac{B_e}{A_e} - \frac{i0}{A_e}. \tag{40}$$

Two active propagators with opposite signs of A_e place their poles on opposite sides of the real axis. If $A_e \sim \delta^{\alpha_e}$ and $D_e \sim \delta^{\kappa_e}$, a leading pole restricts the local coordinate to

$$\Delta\ell^+ \sim \delta^{\kappa_e - \alpha_e}. \tag{41}$$

Propagators for which $\alpha_e + \mathrm{pow}(\Delta\ell^+) > \kappa_e$ depend on ℓ^+ algebraically but are subleading in this local fluctuation and do not generate its leading pinch. This is the precise distinction between a central edge valuation and a loop integration width.

10 Boundary hidden regions

An interior HR has all $x_e > 0$. A boundary HR occurs on a contraction stratum

$$x_{e_1} = \cdots = x_{e_k} = 0. \tag{42}$$

The graph polynomial is first restricted to that stratum, the vanished variables are removed, and the complete analysis is repeated with every remaining active parameter positive. In graph language this is the corresponding contraction minor; in Landau language it is a subleading Landau singularity.

A complete search must include the relevant contraction strata. Failure to find an interior HR is not a proof that the graph has no HR.

11 When do two presentations describe the same HR?

Generator lists are not physical labels. Two presentations describe the same HR when they determine the same physical component of the common pinch locus, the same resolved leading facet, and the same pulled-back total scaling vector. Replacing a generator set by another presentation of the same relevant ideal does not create a new region.

Genuinely distinct HRs can arise from:

- different positive components of the cancellation variety;
- different cancellation ideals or singular hypersurface systems; or
- different lower facets in local coordinates, and hence different certified pullback vectors.

Thus a raw multiplicity of accepted generator presentations must be reduced by ideal, positive-locus and final-facet equivalence before being interpreted as multiple physical HRs.

12 Generator-independent certificates of absence

Factor harvesting is efficient for discovery, but a definitive negative statement should not depend on a maximum factor length, a maximum number of generators or a preferred factorised presentation.

The homogeneity condition (11), together with strict separation from all non-SL monomials, implies that the exponent support of $\mathcal{F}_{\mathrm{SL}}$ is an exposed face of the appropriate starting support. This gives a generator-independent route:

1. enumerate every relevant exposed face of $\mathcal{F}_0$, or of the augmented full support when asymptotic-order alignment is allowed;
2. interpret each face polynomial as a candidate $\mathcal{F}_{\mathrm{SL}}$ selected by its face normal, and discard faces that fail elementary necessary conditions;
3. solve the simultaneous positive logarithmic Landau equations for every surviving candidate $\mathcal{F}_{\mathrm{SL}}$ in $\mathcal{K}$;
4. test each positive solution against $\mathcal{U}$, every occupied restored δ-layer, the hierarchy gap and the resolved lower facet;
5. repeat the analysis on all required contraction strata.

If every face is excluded by one of these exact steps, there is no HR within the stated first-sheet and asymptotic assumptions. If a search limit, timeout or undecided positivity question is reached, the result is *unresolved*, not evidence for absence.

This distinction should be visible at the user interface. An *exploratory profile* may impose finite budgets on the number of candidate generator sets or generator unions and on positivity solvers; it is useful for finding constructive witnesses, but cannot certify absence. A *certified profile* removes every correctness-sensitive combinatorial cap and records whether factor construction, candidate enumeration, positivity tests and exact scaling all completed. A no-HR statement is made only when none of these stages was truncated or timed out. Arbitrary signed two-monomial subsums are not part of the complete construction: the latter uses the irreducible factors of the full derivative and primitive kinematic-channel polynomials. Such monomial-pair tests may be retained as a legacy diagnostic, but not as a completeness criterion.

13 The Hidden Region Finder

The complete conceptual workflow is:

1. **Fix the physical chart.** Specify $\delta \to 0^+$, a common set of real kinematic coordinates $\boldsymbol{s}$, and the inequalities defining the physical domain $\mathcal{K}$. Introduce positive coordinates only locally if they simplify a particular positivity test.

2. **Choose a stratum.** Start in the interior and repeat on the required contraction strata.

3. **Choose the starting face.** Use $\mathcal{F}_0$ in the ordinary case; when necessary enumerate order-aligned exposed faces $\mathcal{F}^{[\phi]}$ of the full δ-dependent polynomial.

4. **Generate candidate cancellation structures.** Apply the mixed-sign test to the individual derivatives of $\mathcal{F}_\star$ only as a necessary prefilter, and harvest primitive non-monomial factors and candidate generator structures. Do not solve the simultaneous derivative equations of $\mathcal{F}_\star$: they generically have no positive solution.

5. **Isolate the superleading support by scaling.** Solve the homogeneous-weight and strict-separation conditions that select $\mathcal{S}_{\rm SL}$ from the starting support. Define $\mathcal{F}_{\rm SL} = \sum_{i\in\mathcal{S}_{\rm SL}} m_i$, determine the candidate relative hierarchy, and decompose $\mathcal{F}_\star = \mathcal{F}_{\rm SL} + \mathcal{F}_{\rm Obs}$. Retain every intermediate occupied layer for the full certification.

6. **Apply the first-sheet pinch test to $\mathcal{F}_{\rm SL}$.** Only after $\mathcal{S}_{\rm SL}$ has been isolated, solve $\partial\mathcal{F}_{\rm SL}/\partial x_e = 0$ simultaneously for all active parameters with $\boldsymbol{x} > 0$ and $\boldsymbol{s} \in \mathcal{K}$. The resulting common positive locus defines the physical cancellation ideal; reject the candidate if this system has no solution.

7. **Resolve all occupied cancellation layers.** Use the ideal filtration of the complete $\mathcal{P} = \mathcal{U} + \mathcal{F}$, positive transverse weights and the actual δ-weight lattice to find the first nonvanishing layer.

8. **Fix and certify the total vector.** Determine $\vec{v}_{\rm HR} = (\boldsymbol{v}_{\rm HR}; 1)$, $W_{\rm SL}$, $W_{\rm HR}$ and the gap. Certify the lower facet in original coordinates when possible; otherwise dissect and pull the facet normal back.

9. **Identify physical equivalence classes.** Merge generator presentations that describe the same positive pinch component and final facet.

10. **State completeness honestly.** A positive HR requires all certification stages. A no-HR conclusion requires exhaustive face and boundary coverage; incomplete searches remain unresolved.

Part II

Examples

14 How to read the colour-coded polynomials

Every example will display the relevant part of $\mathcal{F}$ using one colour convention. The colours refer to weights under the *final total region vector*; in a composite limit this includes both the preliminary asymptotic-order alignment and the relative HRF scaling:

$$\boxed{\begin{aligned} \text{red} &= \text{individually superleading terms at } W_{\mathrm{SL}}, \\ \text{blue} &= \text{resolved leading terms at } W_{\mathrm{HR}}, \\ \text{black} &= \text{all other terms.} \end{aligned}} \tag{43}$$

Thus a conventional facet region contains blue leading terms and black subleading terms, but no red sector. In a hidden region the red monomials have a lower individual weight than the blue monomials, but their sum vanishes on the positive pinch. The blue terms therefore give the actual leading polynomial. Black also includes any intermediate layer that lies in the cancellation ideal and vanishes before the blue layer is reached.

Schematically, the comparison is

$$\mathcal{F}^{(R)} = \delta^{W_R}\mathcal{F}_{W_R} + \mathcal{O}(\delta^{>W_R}), \qquad \text{ordinary facet region,} \tag{44}$$

$$\mathcal{F}^{(\mathrm{HR})} = \delta^{W_{\mathrm{SL}}}\mathcal{F}_{\mathrm{SL}} + \delta^{W_{\mathrm{HR}}}\mathcal{F}_{\mathrm{HR}} + \mathcal{O}(\delta^{>W_{\mathrm{HR}}}), \qquad \text{hidden region,} \tag{45}$$

where the first term in (45) vanishes on the pinch. This is a classification of the terms, not an instruction to discard the red polynomial before the cancellation has been established.

Every family below is presented in the same order:

(i) the kinematic expansion and its physical domain;

(ii) the graph and the assignment of Schwinger parameters;

(iii) the decomposition $\mathcal{F}_\star = \mathcal{F}_{\mathrm{SL}} + \mathcal{F}_{\mathrm{Obs}}$ and its positive cancellation locus;

(iv) the certified total vector $(\boldsymbol{v}; 1)$, including $(W_{\mathrm{SL}}, W_{\mathrm{HR}})$;

(v) the feature that the algorithm must retain in order not to miss the region.

The examples are therefore a tour through possible HR mechanisms, rather than a chronology of how the implementation was developed.

15 The Crown family: an interior seed and its contraction descendants

15.1 Wide-angle kinematics and the three-loop seed

Take four external momenta in the physical $2 \to 2$ domain

$$s_{12} > 0, \qquad s_{23} < 0, \qquad s_{12} > -s_{23}, \tag{46}$$

and regulate the on-shell limit by $p_i^2 = \delta q_i^2$, with fixed nonzero q_i^2 and $\delta \to 0^+$. A common normalization of the q_i^2 is immaterial for the region vector. The Crown has internal edge set

$$\begin{aligned} E_C = \{&(1,5),(1,6),(2,5),(2,6), \\ &(3,5),(3,6),(4,5),(4,6)\}, \end{aligned} \tag{47}$$

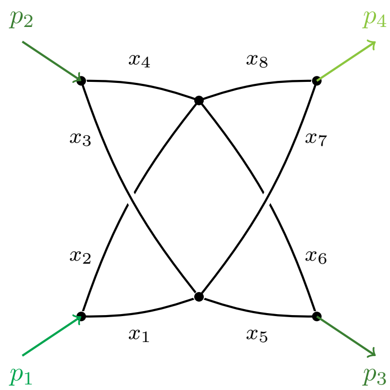

Figure 1: The three-loop Crown with the Schwinger-parameter assignment used in Eq. (47). The external-arrow colours are visual label guides only.

in the order $x_1, \ldots, x_8$, with p_i attached to vertex i for $i = 1, \ldots, 4$. This edge list fixes the graph without relying on a particular drawing.

At fixed parameters the leading polynomial is

$$\begin{aligned}\mathcal{F}_0 = s_{12}(x_4x_5 - x_3x_6)(x_2x_7 - x_1x_8)\\ -s_{23}(x_2x_3 - x_1x_4)(x_6x_7 - x_5x_8).\end{aligned} \tag{48}$$

The four displayed binomials vanish on one common positive component; only three equations are independent there. Hence

$$\mathcal{F}_\star = \mathcal{F}_0 = \mathcal{F}_{\mathrm{SL}}, \qquad \mathcal{F}_{\mathrm{Obs}} = 0. \tag{49}$$

Writing $\mathcal{F} = \mathcal{F}_0 + \delta\mathcal{F}_1 + \cdots$, the certified vector and weights are

$$\vec{v}_{\mathrm{HR}} = (-1,-1,-1,-1,-1,-1,-1,-1;1), \qquad (W_{\mathrm{SL}}, W_{\mathrm{HR}}) = (-4,-3). \tag{50}$$

Consequently

$$\mathcal{P}(\delta^{-1}\boldsymbol{x}, \delta) = \delta^{-4}\mathcal{F}_0 + \delta^{-3}(\mathcal{F}_1 + \mathcal{U}) + \mathcal{O}(\delta^{-2}). \tag{51}$$

The algorithmic lesson is the most elementary one: HRF must keep a simultaneous positive zero of several cancellation factors. No obstruction, non-uniform scaling or preliminary alignment is needed.

15.2 The same seed in Regge limits

Momentum conservation gives $s_{13} = -s_{12} - s_{23}$. The three channel charts used in the audit are

$$\begin{aligned}
T_{23}: &\quad s_{23} = -\delta s_{12}, && s_{12} > 0,\\
T_{12}: &\quad s_{12} = -\delta s_{23}, && s_{23} < 0,\\
T_{13}: &\quad s_{23} = -s_{12} + \delta s_{12}, && s_{12} > 0,
\end{aligned} \tag{52}$$

with $\delta \to 0^+$. The Crown HR is present in all three charts. The surviving cancellation equations are channel dependent, but they are restrictions of the same wide-angle positive pinch. This is the positive seed for the observed wide-angle-to-Regge correspondence.

15.3 SuperCrown and HyperCrown boundary regions

Higher-loop graphs inherit the seed on contraction strata. In edge order, the two graphs are

$$\begin{aligned}
E_{SC} &= \{(1,5),(1,6),(2,7),(2,6),(3,5),(3,8),(4,7),(4,8),(5,7),(6,8),(7,8),(5,6)\},\\
E_{HC} &= \{(1,5),(1,6),(4,9),(4,6),(2,7),(2,6),(3,8),(3,6),(9,5),(7,8),(8,9),(5,7)\}.
\end{aligned} \tag{53}$$

In this subsection the displayed graphs use the one-based order $x_1, \ldots, x_{12}$. The accompanying notebooks use $x_0, \ldots, x_{11}$, so that $x_j^{\text{notebook}} = x_{j+1}^{\text{here}}$. We give both conventions when specifying a boundary; this prevents a contraction label from being confused with a scaling component.

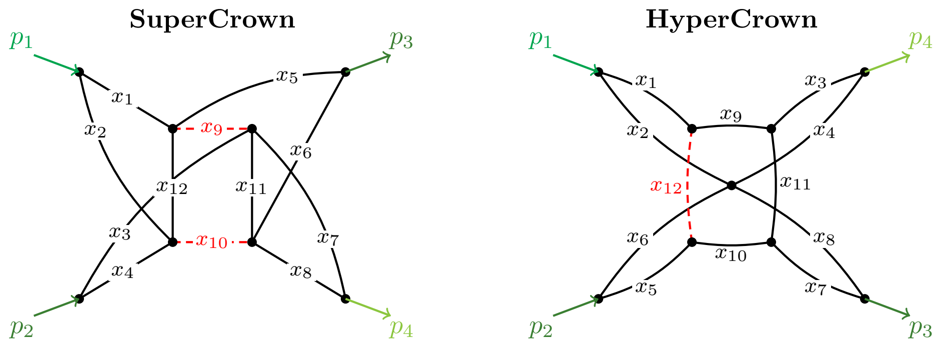

Figure 2: Higher-loop Crown descendants: the five-loop SuperCrown (left) and the four-loop HyperCrown (right). Red dashed edges are set to zero on the representative contraction strata used below: $x_9 = x_{10} = 0$ for the SuperCrown and $x_{12} = 0$ for the HyperCrown, in the one-based convention of the figure.

SuperCrown: a codimension-two inherited Crown

The focused boundary is

$$B_{\text{SC}} = \{x_9 = x_{10} = 0\}_{\text{here}} = \{x_8 = x_9 = 0\}_{\text{notebook}}. \tag{54}$$

After restriction, the surviving superleading polynomial includes the two additional active edge parameters,

$$\mathcal{F}_{\text{SL}}^{\text{SC}} = x_{11}x_{12}\,\mathcal{F}_{\text{SL}}^{C}(x_1, \ldots, x_8), \qquad \mathcal{F}_{\text{Obs}}^{\text{SC}} = 0. \tag{55}$$

Equivalently, its two generators are

$$g_1 = (x_4x_5 - x_3x_6)(x_2x_7 - x_1x_8), \tag{56}$$
$$g_2 = (x_2x_3 - x_1x_4)(x_6x_7 - x_5x_8). \tag{57}$$

In the active order $(x_1, \ldots, x_8, x_{11}, x_{12})$, the certified augmented vector is

$$\vec{v}_{\text{HR}}^{\text{SC}} = (-1, -1, -1, -1, -1, -1, -1, -1, -1, -1; 1), \qquad (W_{\text{SL}}, W_{\text{HR}}) = (-6, -5). \tag{58}$$

The weights include the monomial factor $x_{11}x_{12}$; stripping it would incorrectly quote the Crown weights $(-4, -3)$. The current polynomial-factor run finds one HR on this stratum with no search truncation. The same contraction descendant is present in all three Regge charts.

HyperCrown: two boundary-HR types

The four square edges are

$$Q_{\text{HC}} = \{x_9, x_{10}, x_{11}, x_{12}\}_{\text{here}} = \{x_8, x_9, x_{10}, x_{11}\}_{\text{notebook}}. \tag{59}$$

Their D_4 symmetry acts simultaneously on external, edge and kinematic labels. The wide-angle results through codimension two are

Type	Representative boundary	Certified scaling	Full orbit / conclusion
I	$x_{12} = 0$	the opposite edge x_{11} has $v_{11} = -2$; all other active $v_e = -1$; $(W_{\rm SL}, W_{\rm HR}) = (-6, -5)$	$x_9 = 0$, $x_{10} = 0$, $x_{11} = 0$, $x_{12} = 0$: one HR on each boundary
II	$x_9 = x_{11} = 0$	all ten active components have $v_e = -1$; $(W_{\rm SL}, W_{\rm HR}) = (-5, -4)$	$\{x_9, x_{11}\}$, $\{x_{10}, x_{11}\}$, $\{x_{10}, x_{12}\}$, $\{x_9, x_{12}\}$: one HR on each boundary
Excluded	$x_9 = x_{10} = 0$	no exact coverage vector	the opposite-pair orbit $\{x_9, x_{10}\}$, $\{x_{11}, x_{12}\}$ has no HR

Here every boundary equation means $x_e = 0$, not a value of v_e. Type I decomposes as

$$\mathcal{F}_0^{\rm HC}\big|_{x_{12}=0} = \underbrace{\big[\mathcal{F}_0^{\rm HC}\big]_{W=-6}}_{\mathcal{F}_{\rm SL}^{\rm I}=x_{11}Q_{\rm I}} + \underbrace{\big[\mathcal{F}_0^{\rm HC}\big]_{W=-5}}_{\mathcal{F}_{\rm Obs}^{\rm I}}. \tag{60}$$

The obstruction fixes the extra unit of scaling on the opposite square edge. For Type II every surviving term has weight -5, so

$$\mathcal{F}_0^{\rm HC}\big|_{x_9=x_{11}=0} = \mathcal{F}_{\rm SL}^{\rm II}, \qquad \mathcal{F}_{\rm Obs}^{\rm II} = 0. \tag{61}$$

Two accepted generator presentations on a direct Type-II representative give the same positive component and vector, hence one physical HR. Momentum reconstruction identifies Type I as the four-jet Landshoff flow plus one soft loop, and Type II as a Landshoff descendant plus one external-direction collinear loop; neither requires a generic-angle Glauber mode.

The opposite-pair orbit is excluded by an uncapped complete-polynomial-factor and exact-coverage audit. It has one s_{12}-supported generator and no s_{23} companion. By contrast, the HyperCrown interior remains unresolved because the available search was capped and one trial timed out. Thus symmetry may complete a positive orbit, while absence requires every search budget to be exposed and exhausted.

The converse audit completes the tested four-loop picture. Sixteen mixed-sign topologies without a Crown contraction minor have no certified interior or boundary HR at wide angle or in any of the three Regge channels. Thus the tested sample supports both the Crown-minor conjecture and the wide-angle-to-Regge conjecture through four loops. These are empirical statements for the audited graphs, not general graph-theoretic theorems.

16 One five-point seed in two expansions: spacelike collinear and central-soft MRK

The next two applications use the same unlabeled two-loop six-propagator topology but different external attachments and different kinematic expansions. The first is the spacelike-collinear limit for which this seed was originally identified. The second is a composite five-point multi-Regge limit in which the central emission becomes soft while the rapidity gaps remain large. The cancellation ideals, total vectors and momentum modes are different, but in both cases the same sparse topology supports the HR. This is the five-point counterpart of the appearance of the Crown in both wide-angle and Regge expansions.

16.1 Spacelike-collinear kinematics and the two-loop seed graph

The collinear chart is defined by

$$\begin{gathered} s_{12} = sz, \qquad s_{23} = -4\delta^2, \qquad s_{34} = -s\chi(z-1), \\ s_{45} = s, \qquad s_{15} = s\chi + c\delta, \qquad \delta \to 0^+, \end{gathered} \tag{62}$$

with

$$s > 0, \qquad z > 1, \qquad -1 < \chi < 0, \tag{63}$$

and fixed nonzero c. Keeping χ negative is useful: the same variables then display directly the change from timelike to spacelike collinear kinematics.

The two-loop seed has edge set

$$E_5 = \{(1,3),(1,5),(2,3),(2,5),(3,4),(4,5)\} \longleftrightarrow (x_1,\dots,x_6), \tag{64}$$

and external attachment $p_i \to i$, $i = 1,\dots,5$. We use $x_{i,j,\dots} = x_i x_j \cdots$.

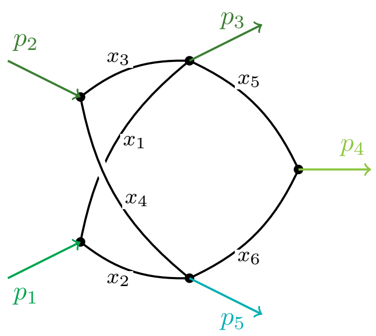

Figure 3: The two-loop five-point seed graph, redrawn in the style of Ref. [5], with the Schwinger parameters used in Eqs. (66)–(70).

16.2 Spacelike collinear: one polynomial, an ordinary facet and a hidden region

For the ordinary facet vector

$$\vec{v}_C = (-2, 0, -2, -2, -2, 0; 1), \tag{65}$$

the expanded polynomial is classified as

$$\mathcal{F}_0^{[C]} = s\big(x_{2,5} + \chi x_{1,6}\big)\big[(z-1)x_3 - x_4\big] + sz(\chi+1)x_{2,3,6}, \tag{66}$$

$$\mathcal{F}_\delta^{[C]} = -c\delta x_{1,4,6} + c\delta x_{2,3,6} + 4\delta^2 x_{1,4,5} - 4\delta^2 x_{2,3,5}. \tag{67}$$

There is no red sector: the blue monomials form an ordinary lower facet.

For the hidden-region vector

$$\vec{v}_{\mathrm{HR}} = (-2,-1,-2,-2,-2,-1;1), \qquad (W_{\mathrm{SL}}, W_{\mathrm{HR}}) = (-5,-4), \tag{68}$$

the identical polynomial instead reads

$$\mathcal{F}_0^{[\mathrm{HR}]} = \underbrace{s\big(x_{2,5} + \chi x_{1,6}\big)\big[(z-1)x_3 - x_4\big]}_{\mathcal{F}_{\mathrm{SL}}} + \underbrace{sz(\chi+1)x_{2,3,6}}_{\mathcal{F}_{\mathrm{Obs}}}, \tag{69}$$

$$\mathcal{F}_\delta^{[\mathrm{HR}]} = -c\delta x_{1,4,6} + c\delta x_{2,3,6} + 4\delta^2 x_{1,4,5} - 4\delta^2 x_{2,3,5}. \tag{70}$$

The cancellation ideal is generated by

$$f_1 = x_{2,5} + \chi x_{1,6}, \qquad f_2 = (z-1)x_3 - x_4, \qquad \mathcal{I} = \langle f_1, f_2 \rangle. \tag{71}$$

The red product vanishes at the positive pinch. The blue obstruction, the blue kinematically suppressed terms, and

$$\mathcal{U}_{W_{\mathrm{HR}}} = x_{1,3} + x_{1,4} + x_{1,5} + x_{3,5} + x_{4,5} \tag{72}$$

form the physical layer. This example introduces the obstruction: $\mathcal{F}_{\mathrm{Obs}} \neq 0$ fixes a non-uniform scaling that would not be determined by the factorized cancellation sector alone.

Complete momentum flow

Table 1 gives the edge assignment for both the ordinary facet and the HR. The momentum q_e flows through the edge carrying x_e, and each triple denotes the exponents of $(q_e^+, q_e^-, |q_{e\perp}|)$. The last column in each case is the power κ_e of q_e^2.

	facet $\vec{v}_C$		hidden $\vec{v}_{\mathrm{HR}}$	
edge	(a_e, b_e, c_e)	κ_e	(a_e, b_e, c_e)	κ_e
q_1	$(0,2,1)$	2	$(1,1,1)$	2
q_2	$(0,0,1)$	0	$(1,0,1)$	1
q_3	$(0,2,1)$	2	$(0,2,1)$	2
q_4	$(0,2,1)$	2	$(0,2,1)$	2
q_5	$(0,2,1)$	2	$(1,1,1)$	2
q_6	$(0,0,0)$	0	$(0,0,0)$	1

Table 1: Complete edge-momentum valuations for the five-point spacelike-collinear facet and HR. Edge q_e corresponds to parameter x_e in Fig. 3.

In the facet region the table admits a homogeneous collinear loop basis and no transverse-dominated loop rerouting. In the HR, q_1 and q_5 are separately soft. Vertex conservation forces the minus components to cancel in the independent rerouting

$$\ell_G = q_1 - q_5 \sim (\delta, \delta^2, \delta), \tag{73}$$

while one may take $\ell_s = q_1 \sim (\delta, \delta, \delta)$ as the other loop variable. Thus the HR has one soft and one Glauber loop. The entry for q_6 is cancellation enhanced: its leading longitudinal and transverse terms cancel so that $q_6^2 \sim \delta$, despite all three components being of order one. This coefficient-level structure is known explicitly in the spacelike-collinear solution, so this reconstruction is mode complete.

Parameter- and momentum-space power counting

For unit propagator powers, the scalar LP representation contains $\mathcal{P}^{-D/2}$. The edge-vector sum is $\sum_e v_e = -10$. Only a restricted neighbourhood of the cancellation locus contributes: in local coordinates one cancellation-normal variable has one power less support than a generic variable with the same edge scaling. The parameter measure is therefore

$$\delta^{-10}\,\delta^{+1} = \delta^{-9}. \tag{74}$$

Since the resolved LP polynomial has weight -4, $\mathcal{P}^{-D/2} \sim \delta^{2D}$, and hence

$$I_{\mathrm{SC}}^{\mathrm{scalar}} \sim \delta^{-9+2D} = \delta^{-1-4\epsilon}, \qquad D = 4 - 2\epsilon. \tag{75}$$

The power enhancement is a property of the unit-numerator scalar integral. For the gauge-theory numerator relevant to the amplitude, one additional power of δ gives the leading-power behaviour $\delta^{-4\epsilon}$.

The same count is obtained in momentum space. A convenient routing contains one soft loop and one Glauber loop,

$$\ell_s \sim (\delta, \delta, \delta), \qquad \ell_G \sim (\delta, \delta^2, \delta), \tag{76}$$

where the entries denote $(q^+, q^-, |q_\perp|)$. Their measures scale as δ^D and δ^{D+1}, respectively. The six propagator virtualities have powers

$$(2, 1, 2, 2, 2, 1), \tag{77}$$

whose sum is ten. Therefore

$$I^{\text{scalar}}_{\text{SC}} \sim \delta^{D+(D+1)-10} = \delta^{-1-4\epsilon}, \tag{78}$$

in agreement with Eq. (75). In the original edge routing the Glauber momentum is the difference of two soft edge momenta. This is why inspection of individual propagator modes alone does not reveal it.

16.3 The vertex-corrected graph: a genuinely non-binomial generator

The same kinematics applied to the three-loop vertex-corrected graph gives

$$\begin{aligned} E_{5V} = \{&(1,3),(1,5),(3,7),(6,5),(3,4),\\ &(4,5),(2,7),(6,7),(2,6)\} \longleftrightarrow (x_0,\dots,x_8). \end{aligned} \tag{79}$$

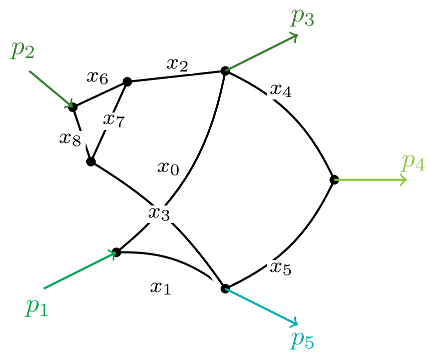

Figure 4: The three-loop vertex-corrected five-point graph, drawn as the two-loop seed of Fig. 3 with a local vertex correction adjacent to p_2. In particular, x_7 lies outside the other loops. The labels $x_0,\dots,x_8$ are those used in Eqs. (80)–(82).

Here a pairwise monomial search is insufficient. A convenient primitive generator is $g_1 g_2$, where

$$\begin{aligned} g_1 &= (z-1)(x_2x_6 + x_2x_7 + x_2x_8) + z x_6 x_7 \\ &\quad - (x_3x_6 + x_3x_7 + x_3x_8 + x_6x_7 + x_7x_8), \\ g_2 &= x_1x_4 + \chi x_0 x_5. \end{aligned} \tag{80}$$

The first factor has more than two monomials and cannot be represented by a single equality of a monomial pair. The decomposition is nevertheless as simple at ideal level as in the seed:

$$\mathcal{F}_\star = \underbrace{-s\, g_1 g_2}_{\mathcal{F}_{\text{SL}}} + \underbrace{-s(1+\chi) z\, x_1 x_5 (x_2x_6 + x_2x_7 + x_6x_7 + x_2x_8)}_{\mathcal{F}_{\text{Obs}}}, \tag{81}$$

$$\vec{v}_{\text{HR}} = (-2,-1,-2,-2,-2,-1,-2,-2,-2;1), \qquad (W_{\text{SL}}, W_{\text{HR}}) = (-7,-6). \tag{82}$$

The lesson is not merely that long factors may occur. HRF must construct and test the ideal generated by primitive polynomial factors, allow polynomial multipliers in $\mathcal{F}_{\text{SL}}$, and solve the positive pinch only after the scaling law has isolated that sector. A binomial-only or pair-sector-only candidate list would miss this region.

16.4 Central-soft five-point multi-Regge kinematics

We now keep the multi-Regge rapidity ordering but approach the additional surface on which the central emitted momentum becomes soft. Use the all-outgoing convention

$$-p_1 - p_2 + p_3 + p_4 + p_5 = 0, \tag{83}$$

so that p_1, p_2 are incoming and p_3, p_4, p_5 are ordered from forward to backward rapidity. Let $0 < a < b$ and choose the exact on-shell chart

$$\begin{aligned}
p_3^+ &= P\delta^{-b}, & p_3^- &= \frac{T}{P}\delta^b, & \boldsymbol{p}_{3\perp} &= \boldsymbol{q}, \\
p_4^+ &= K\delta^a, & p_4^- &= \frac{R}{K}\delta^a, & \boldsymbol{p}_{4\perp} &= \delta^a \boldsymbol{k}, \\
p_5^- &= M\delta^{-b}, & p_5^+ &= \frac{T + \sigma C\delta^a + R\delta^{2a}}{M}\delta^b, & \boldsymbol{p}_{5\perp} &= -\boldsymbol{q} - \delta^a \boldsymbol{k},
\end{aligned} \tag{84}$$

where

$$T = |\boldsymbol{q}|^2, \qquad R = |\boldsymbol{k}|^2, \qquad 2\,\mathrm{Re}(\boldsymbol{q}\bar{\boldsymbol{k}}) = \sigma C, \qquad C^2 < 4RT, \tag{85}$$

and $P, M, K, R, T, C > 0$, $\sigma = \pm 1$. The incoming components are fixed by Eq. (83), so on-shellness and momentum conservation hold at every nonzero δ, not only at leading power.

The rapidities obey

$$y_3 \sim -b \log \delta, \qquad y_4 = O(1), \qquad y_5 \sim b \log \delta, \tag{86}$$

while all components of p_4 scale as δ^a. Thus the rapidity hierarchy of MRK is preserved, but the usual assumption of comparable transverse momenta is not: this is a composite central-soft MRK limit. For the representative path $(a, b) = (1, 2)$,

$$\begin{aligned}
s_{12} &= PM\delta^{-4} + O(\delta^{-1}), & s_{34} &= \frac{PR}{K}\delta^{-1} + O(\delta), & s_{45} &= KM\delta^{-1} + O(\delta), \\
s_{23} &= -T + O(\delta^3), & s_{15} &= -T - \sigma C\delta - R\delta^2 + O(\delta^3). &&
\end{aligned} \tag{87}$$

This has the physical signs $s_{12}, s_{34}, s_{45} > 0$ and $s_{23}, s_{15} < 0$. In ordinary five-point MRK the central transverse momentum remains finite; the HR described below is absent in that expansion.

16.5 Central-soft MRK certificate for the seed

The internal edges are again those of Eq. (64). To match the MRK computation we relabel them as

$$\{(1,3), (1,5), (2,3), (2,5), (3,4), (4,5)\} \longleftrightarrow (x_0, \ldots, x_5), \tag{88}$$

and use the representative attachment

$$p_1 \to 1, \quad p_2 \to 2, \quad p_3 \to 3, \quad p_5 \to 4, \quad p_4 \to 5. \tag{89}$$

Relative to Fig. 3, the last two external labels are interchanged: the central gluon is attached to a four-valent internal vertex. A scan of the twelve inequivalent external attachments finds an HR for precisely the six in which p_4 is attached to either of the two four-valent vertices. All twelve one-loop pentagons are negative, and the same two-loop attachments are negative in generic MRK. The result is unchanged for $(a, b) = (1, 2), (1, 3), (2, 3)$ and for both signs σ.

For Eq. (89), asymptotic-order alignment gives

$$\boldsymbol{\phi} = (-3, -3, 0, -3, -3, -3). \tag{90}$$

On the exposed face the superleading polynomial factorises as

$$\mathcal{F}_\star^{\mathrm{MRK}} = KM^2P\,(Kx_3 - Px_2)(x_1x_4 - x_0x_5), \qquad \mathcal{F}_{\mathrm{Obs}} = 0, \tag{91}$$

up to an irrelevant overall sign. Its positive cancellation locus is

$$Px_2 - Kx_3 = 0, \qquad x_1x_4 - x_0x_5 = 0. \tag{92}$$

The second-stage HRF vector in the aligned coordinates is uniform,

$$\boldsymbol{\rho} = (-1, -1, -1, -1, -1, -1), \tag{93}$$

and Eq. (25) gives the total vector

$$\vec{v}_{\mathrm{HR}}^{\mathrm{MRK}} = (-4, -4, -1, -4, -4, -4; 1). \tag{94}$$

The final aligned HRF step has gap one. For absolute counting in the original exact chart, however, the raw superleading layer has weight -13 and the resolved scaleful LP layer has weight -8. The full cancellation depth is therefore five: four powers are exposed by alignment and one by the final HRF step. Quoting only the latter would lose essential information.

Complete edge flow and mode-adapted loop basis

The full valuation-level solution is shown in Table 2. Here q_e denotes the momentum on the edge carrying x_e in Eq. (88). "Enhanced" means that the displayed components would give a lower virtuality power unless their leading longitudinal and transverse contributions cancel.

edge	(a_e, b_e, c_e)	q_e^2	realization
q_0	$(6, -2, 2)$	δ^4	balanced
q_1	$(6, -2, 2)$	δ^4	balanced
q_2	$(-2, 3, 2)$	δ	longitudinal dominated
q_3	$(1, 3, 2)$	δ^4	balanced
q_4	$(2, -2, 0)$	δ^4	enhanced
q_5	$(4, -2, 1)$	δ^4	enhanced

Table 2: Complete edge-momentum valuations for the central-soft MRK HR. The table satisfies the target virtualities and tropical vertex conservation. The pole analysis below supplies the additional local widths required for a mode-complete solution.

The edge momenta q_0 and q_3 form one convenient chord basis, but their central edge powers are not the local widths of a mode-adapted basis. Set

$$r = q_0, \qquad \ell = q_0 - q_4. \tag{95}$$

Momentum conservation then gives

$$q_2 = p_3 - \ell, \qquad q_3 = \ell + p_2 - p_3, \qquad q_4 = r - \ell, \qquad q_5 = p_5 - r + \ell. \tag{96}$$

The central value is

$$\ell\big|_{\mathrm{pinch}} \sim (\delta^2, \delta^2, 1), \tag{97}$$

but its local integration widths are more restrictive.

Four denominators depend algebraically on ℓ^+. For q_2 and q_3, however, the coefficient is of order δ^3. A local $\Delta\ell^+ \sim \delta^6$ fluctuation enters them only at order δ^9, subleading relative to their virtualities δ and δ^4. The two leading-sensitive denominators are instead

$$D_4 = (r^+ - \ell^+)(r^- - \ell^-) - |r_\perp - \ell_\perp|^2 + i0, \tag{98}$$

$$D_5 = (p_5^+ - r^+ + \ell^+)(p_5^- - r^- + \ell^-) - |p_{5\perp} - r_\perp + \ell_\perp|^2 + i0. \tag{99}$$

Writing $q_4^- = r^- - \ell^-$ and $q_5^- = p_5^- - r^- + \ell^-$, their poles are

$$\ell_4^+ = r^+ - \frac{|q_{4\perp}|^2}{q_4^-} + \frac{i0}{q_4^-}, \tag{100}$$

$$\ell_5^+ = r^+ - p_5^+ + \frac{|q_{5\perp}|^2}{q_5^-} - \frac{i0}{q_5^-}. \tag{101}$$

On the physical positive-flow branch $q_4^-, q_5^- > 0$, the first pole is above and the second below the real axis. Both slopes have power -2, whereas $D_4, D_5 \sim \delta^4$. Equations (40)–(41) therefore give

$$\Delta\ell^+ \sim \delta^6. \tag{102}$$

The residual relation $q_2 = p_3 - \ell$ fixes $\Delta\ell^- \sim \delta^3$ and $\Delta\ell_\perp \sim \delta^2$. Thus

$$\ell : \quad (\Delta\ell^+, \Delta\ell^-, |\Delta\ell_\perp|) \sim (\delta^6, \delta^3, \delta^2). \tag{103}$$

Since $6 + 3 > 2 \times 2$, ℓ is one genuine Glauber integration mode. There are not two Glauber loops. The other independent variable has local widths $r \sim (\delta^6, \delta^{-2}, \delta^2)$. Its plus-component width is independently confirmed by the q_0 and $q_1 = p_1 - r$ poles, which also approach from opposite half-planes and give $\Delta r^+ \sim \delta^6$.

Parameter- and momentum-space power counting

The sum of the six edge exponents in Eq. (94) is -21. Restriction to the simultaneous cancellation neighbourhood restores five powers, giving the effective parameter-measure weight -16. Since the resolved LP weight is -8,

$$I^{\text{scalar}}_{\text{MRK}} \sim \delta^{-21+5}\big(\delta^{-8}\big)^{-D/2} = \delta^{4D-16} = \delta^{-8\epsilon}. \tag{104}$$

The unit-numerator scalar HR is therefore leading power at $D = 4$, with the nontrivial ϵ-dependence expected of an infrared region.

The pole analysis now gives an independent momentum-space count. The r measure has power $6 - 2 + 2(D-2) = 2D$, while the Glauber measure has power $6 + 3 + 2(D-2) = 2D + 5$. The virtuality powers in Table 2 sum to 21. Hence

$$\frac{\mathrm{d}^D\ell_1\, \mathrm{d}^D\ell_2}{\prod_{e=0}^{5} q_e^2} \sim \delta^{,2D+(2D+5)-21} = \delta^{4D-16} = \delta^{-8\epsilon}. \tag{105}$$

This agrees with Eq. (104). No separate five-power cancellation-depth correction is inserted in momentum space: the restricted support is already encoded in the local widths, most visibly in $\Delta\ell^+ \sim \delta^6$. Treating the central powers of (q_0, q_3) as integration widths and then adding the cancellation depth would double count the restriction.

16.6 Comparison: one seed, two different hidden regions

Expansion	total edge vector	scalar power	momentum signature
Spacelike collinear	$(-2, -1, -2, -2, -2, -1; 1)$	$\delta^{-1-4\epsilon}$	soft + Glauber
Central-soft MRK	$(-4, -4, -1, -4, -4, -4; 1)$	$\delta^{-8\epsilon}$	one genuine Glauber loop

These are not two coordinate descriptions of one region. Their kinematic limits, external attachments, cancellation ideals, vectors and scalar powers are different. What they share is the same unlabeled two-loop seed. The comparison suggests a broader organisational principle: hidden regions are sparse because only special graph topologies support the required positive pinches, but once such a seed exists it can reappear through distinct cancellation mechanisms in more than one kinematic expansion. The Crown in wide-angle and Regge limits is the four-point example; the present seed in spacelike-collinear and central-soft MRK limits is the five-point example. The six-point NMRK/DSC pair below supplies a third instance, although its topological interpretation remains to be completed.

17 One hexagon, two composite limits: NMRK with $w = z$ and DSC

17.1 Common graph and why $\mathcal{F}_0$ is insufficient

Both examples use the one-loop hexagon

$$\begin{gathered}(5,6),(6,1),(1,2),(2,3),(3,4),(4,5) \longleftrightarrow x_0,x_1,x_2,x_3,x_4,x_5,\\ p_1 \to 1, \quad p_2 \to 4, \quad p_3 \to 5, \quad p_4 \to 3, \quad p_5 \to 2, \quad p_6 \to 6,\end{gathered} \tag{106}$$

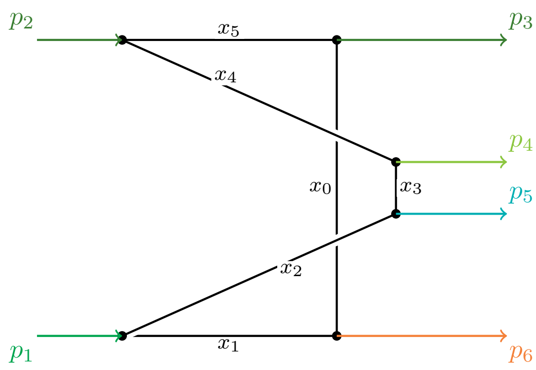

Figure 5: The common one-loop hexagon used for the NMRK and DSC analyses. The nearly straight upper chain $p_2 - x_5 - p_3$ and lower chain $p_1 - x_1 - p_6$ display the two spacelike-collinear pairs. On the right, the vertical separations display the rapidity ordering $p_3 \gg (p_4, p_5) \gg p_6$; the relative order of p_4, p_5 is immaterial in NMRK. The two crossings are deliberate and are not additional vertices. This physical attachment ordering supports the HR (and associated Regge cut), in contrast to the genuinely planar ordering 123456 with all four emissions on one side.

with $\mathcal{U} = x_0 + x_1 + x_2 + x_3 + x_4 + x_5$. The full graph polynomial is the same in both limits; only the invariant substitution changes.

In each case the cancelling monomials occur at different fixed-parameter orders. Thus $\mathcal{F}_0$ does not contain the relevant cancellation sector. A face vector f first performs asymptotic-order alignment and selects $\mathcal{F}_\star = \mathcal{F}^{[f]}$. Because f and $f + c\mathbf{1}$ select the same face, this discovery vector is not yet the physical region vector. The full occupied layer structure and $\mathcal{U}$ must fix the uniform component.

17.2 Two-loop topology test and possible Regge-cut interpretation

The two nearly horizontal chains in Fig. 5 are the (23) and (16) collinear sectors. A t-channel exchange is an internal path connecting these two chains. The two central emissions p_4, p_5 may occur on the same such path or on two separate paths. This distinction becomes explicit in the two-loop graphs used in the NMRK and DSC audits. With the external attachments of Eq. (106), their internal edges, in $x_0, \ldots, x_8$ order, are

$$\begin{aligned} E_{\mathrm{PH}} &= \{(5,6),(1,8),(8,6),(1,2),(2,3),(3,4),(4,7),(7,5),(7,8)\},\\ E_{\mathrm{NPH}} &= \{(5,8),(1,8),(1,2),(2,3),(7,3),(7,5),(4,7),(6,8),(4,6)\},\\ E_{\mathrm{HP}} &= \{(5,6),(1,8),(8,6),(1,3),(3,4),(4,7),(7,5),(7,2),(8,2)\}. \end{aligned} \tag{107}$$

The current certificates give

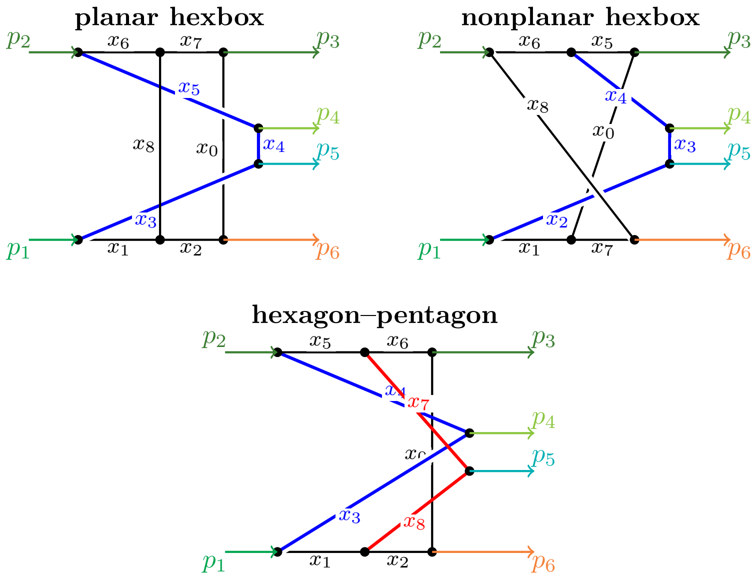

Figure 6: The two-loop graphs in the same rapidity-ordered embedding as Fig. 5. The upper and lower horizontal chains are the (23) and (16) collinear sectors. In each hexbox, the single blue t-channel path carries both central emissions. In the hexagon–pentagon, p_4 lies on the blue path and p_5 on the distinct red path. Crossings without black dots are not vertices.

Graph	central-emission paths	NMRK with $w = z$	generic DSC
one-loop hexagon	same	1 full certificate	1 full certificate
planar hexbox	same	7 full + 4 boundary	1 full certificate
nonplanar hexbox	same	1 full + 2 boundary	no certificate
hexagon–pentagon	separate	no staged candidate	no staged candidate

These entries count certified vectors, not necessarily distinct physical regions. Within the tested NMRK set, however, the same-exchange graphs have an HR and the separate-exchange graph does not. This is evidence for a topology criterion, not yet a theorem; the DSC nonplanar-hexbox result also shows that the topology is not sufficient in every kinematic expansion.

The crossed ordering is the six-point Mandelstam region in which Regge-cut effects arise. Bartels' reggeon calculus organized high-energy amplitudes by one- and two-reggeon states in successive t-channels [6]. Bartels, Lipatov and Sabio Vera later showed that, for six and more legs, the BDS ansatz is incompatible with the Regge cuts predicted by the BFKL approach in appropriate physical channels [7], and computed the leading-logarithmic color-octet cut contribution and the corresponding $2 \to 4$ and $3 \to 3$ discontinuities [8]. That correction is the high-energy contribution that must reside in the six-point BDS remainder function, and it initiated the extensive subsequent study of remainder functions in multi-Regge kinematics.

It is therefore natural to conjecture that the HR found here is the parameter-space origin of the relevant t-channel cut. This connection is not established by the present certificates alone. It requires matching the local positive pinch and its region integral to the appropriate Mandelstam discontinuity and multi-reggeon exchange. Enlarging the topology sample in Fig. 6 is a direct way to test the conjecture.

17.3 NMRK with the additional surface $w = z$

Use the MSLCV chart

$$X_{34} = \frac{\widehat{X}_{34}}{\eta}, \qquad X_{56} = \frac{\widehat{X}_{56}}{\eta}, \qquad X_{45}, Q, z, \bar{z}, w, \bar{w} = O(1), \qquad \eta \to 0^+, \tag{108}$$

with

$$Q, \widehat{X}_{34}, X_{45}, \widehat{X}_{56}, K_z > 0, \qquad K_z = (1-z)(1-\bar{z}) > 0. \tag{109}$$

On this physical NMRK sheet $z, \bar{z}$ and $w, \bar{w}$ are conjugate pairs. The restriction $w = z$, $\bar{w} = \bar{z}$ is additional to NMRK. For generic NMRK the relevant bilinear determinant is proportional to $(w-z)(\bar{w}-\bar{z})$, so the HR is absent.

On the restricted surface define

$$L_1^N = (1+X_{45})x_4 - \widehat{X}_{34}X_{45}x_5, \qquad L_2^N = (1+X_{45})x_2 - K_z X_{45}\widehat{X}_{56}x_1. \tag{110}$$

The selected face has

$$\mathcal{F}_\star^N = \mathcal{F}_{\mathrm{SL}}^N + \mathcal{F}_{\mathrm{Obs}}^N = Q\widehat{X}_{34}\widehat{X}_{56}L_1^N L_2^N + 0. \tag{111}$$

The two-stage vectors and certified answer are

$$\begin{aligned} \boldsymbol{\phi}_N &= (-1,-1,-2,-1,-2,-1), & \boldsymbol{\rho}_N &= (-1,-1,-1,-1,-1,-1),\\ \vec{v}_{\mathrm{HR}}^N &= (-2,-2,-3,-2,-3,-2;1), & (W_{\mathrm{SL}}, W_{\mathrm{HR}}) &= (-4,-3). \end{aligned} \tag{112}$$

There is no intermediate cancellation layer. At W_{HR},

$$\begin{aligned} \mathcal{U}_{-3} &= x_2 + x_4,\\ \mathcal{F}_{-3} &= -Q\widehat{X}_{34}X_{45}\widehat{X}_{56}(K_z x_0 x_2 + K_z x_0 x_4 + \widehat{X}_{34}x_3x_5 + K_z\widehat{X}_{56}x_1x_3). \end{aligned} \tag{113}$$

These monomials give a unique lower facet directly. The required new capability is asymptotic-order alignment itself; once the correct face is exposed, the cancellation has depth one.

17.4 Generic double spacelike-collinear kinematics

For DSC let $\varepsilon \to 0^+$ and use the real physical slice

$$a < 0, \qquad -1 < \tau_1 < 0, \qquad \tau_2 < -1, \qquad z_D < 0, \qquad \bar{z}_D < 0. \tag{114}$$

The symbols $z_D, \bar{z}_D$ are conjugate on the standard real transverse MRK branch, but after continuation to this DSC sheet they are two independent real roots. In the graph-polynomial sign convention the adjacent invariants behave as

$$\begin{aligned} s_{12} &= -\frac{\tau_2}{a(1+\tau_1)(1+\tau_2)} + O(\varepsilon), & s_{23} &= \frac{\varepsilon^2\tau_1}{a\bar{z}_D} + O(\varepsilon^3),\\ s_{34} &= \frac{1}{(a-1)(1+\tau_2)} + O(\varepsilon), & s_{45} &= -\frac{1}{a},\\ s_{56} &= \frac{\tau_1}{(a-1)(1+\tau_1)} + O(\varepsilon), & s_{16} &= \frac{\varepsilon^2\tau_2 z_D}{a} + O(\varepsilon^3). \end{aligned} \tag{115}$$

Thus $s_{23}, s_{16} < 0$, while $s_{12}, s_{34}, s_{45}, s_{56} > 0$, as required for $2 \to 4$ scattering with legs 1 and 2 incoming.

The cancellation factors and selected-face decomposition are

$$L_1^D = (1+\tau_2)x_4 + x_5, \qquad L_2^D = \tau_1 x_1 + (1+\tau_1)x_2, \tag{116}$$

$$\mathcal{F}_\star^D = \mathcal{F}_{\mathrm{SL}}^D + \mathcal{F}_{\mathrm{Obs}}^D = -(a-1)a^2\bar{z}_D^2 L_1^D L_2^D + 0. \tag{117}$$

Both factors have positive zeros,

$$x_5 = -(1+\tau_2)x_4 > 0, \qquad x_2 = -\frac{\tau_1}{1+\tau_1}x_1 > 0. \tag{118}$$

For crossed double-timelike collinear splittings $\tau_i > 0$, so these same factors have one sign for positive x_i and the pinch disappears.

The alignment vector and the provisional face-only HRF vector are

$$\boldsymbol{\phi}_D = (-2, -2, -2, -1, -2, -2), \qquad \boldsymbol{\rho}_D^{(0)} = (-1, -1, -1, -1, -1, -1). \tag{119}$$

The provisional vector is obtained from the isolated aligned face; it is not the second vector in the final composition. With $\mathcal{I}_D = \langle L_1^D, L_2^D \rangle$, the actual occupied layers obey

$$\underbrace{\mathcal{F}_{-4} \in \mathcal{I}_D^2}_{\mathrm{ord}_{\mathcal{I}_D}=2} + \underbrace{\mathcal{F}_{-3} \in \mathcal{I}_D \setminus \mathcal{I}_D^2}_{\mathrm{ord}_{\mathcal{I}_D}=1} + \underbrace{\mathcal{F}_{-2} \notin \mathcal{I}_D}_{\mathrm{ord}_{\mathcal{I}_D}=0}. \tag{120}$$

The middle layer contains 52 nonzero terms. It is not a missing power of ε: it vanishes to first order on the common pinch. The leading $\mathcal{U}$ layer is

$$\mathcal{U}_{-2} = x_0 + x_1 + x_2 + x_4 + x_5, \tag{121}$$

and the full ideal-jet balance gives

$$\begin{aligned} \boldsymbol{\rho}_D &= (0, 0, 0, 1, 0, 0), & \boldsymbol{\phi}_D + \boldsymbol{\rho}_D &= (-2, -2, -2, 0, -2, -2), \\ \vec{v}_{\mathrm{HR}}^{D} &= (-2, -2, -2, 0, -2, -2; 1), & (W_{\mathrm{SL}}, W_{\mathrm{HR}}) &= (-4, -2), \end{aligned} \tag{122}$$

with gap two. A local dissection pulls back to this same vector.

This example is why asymptotic-order alignment cannot be treated as a cosmetic preprocessing step. HRF must (i) expose the cancelling face, (ii) retain every occupied layer, (iii) compute successive ideal orders, and (iv) use the first nonvanishing LP layer to fix the uniform ambiguity of the face representative. Simply adding $\boldsymbol{\phi}_D + \boldsymbol{\rho}_D^{(0)}$ would give the wrong total vector and the wrong cancellation depth. This is the concrete example for the distinction between $\boldsymbol{\rho}^{(0)}$ and the corrected $\boldsymbol{\rho}$ in Sec. 6.

17.5 What the two limits share—and what they do not

After their separate kinematic rescalings, the factor coefficients are identified by

$$\frac{K_z X_{45} \widehat{X}_{56}}{1 + X_{45}} = -\frac{\tau_1}{1 + \tau_1}, \qquad \frac{\widehat{X}_{34} X_{45}}{1 + X_{45}} = -\frac{1}{1 + \tau_2}. \tag{123}$$

Hence the two order-aligned $\mathcal{F}_{\mathrm{SL}}$ supports describe the same type of positive cancellation hypersurface. The next layers differ:

Expansion	certified $\boldsymbol{v}$	$(W_{\mathrm{SL}}, W_{\mathrm{HR}})$	depth
NMRK with $w = z$	$(-2, -2, -3, -2, -3, -2)$	$(-4, -3)$	1
generic DSC	$(-2, -2, -2, 0, -2, -2)$	$(-4, -2)$	2

The common generator does not make the two limits identical. The first resolved LP layer, and therefore the physical region vector, remembers the path by which the singular hypersurface is approached.

18 What the examples teach

Family	New structure	Capability needed not to miss the HR
Crown	simultaneous positive cancellation, $\mathcal{F}_{\mathrm{Obs}} = 0$	keep a multi-equation positive pinch and determine its uniform scaling
Crown descendants	contraction-minor boundary regions	scan boundary strata and compare restricted ideals, not only parent interiors
Five-point seed	$\mathcal{F}_{\mathrm{Obs}} \neq 0$, non-uniform scaling	isolate $\mathcal{F}_{\mathrm{SL}}$ before imposing the positive pinch; retain the obstruction
Vertex correction	primitive non-binomial factor	generate and test general polynomial ideals rather than monomial pairs only
Five-point central-soft MRK	cross-order cancellation and a candidate Glauber rerouting	compose the two vectors and distinguish edge values from loop fluctuation widths
NMRK, $w = z$	cancellation on a nontrivial asymptotic face	align orders before running the ordinary HRF construction
DSC	a second occupied cancellation layer	resolve the ideal filtration and fix the uniform face ambiguity from the full LP polynomial

The progression is cumulative. Later examples require the earlier tests in addition to their new feature. In every case the final statement is again geometric: after cancellation is resolved, the leading polynomial must be a lower facet in coordinates adapted to the pinch. HRF reconstructs the coefficient and Landau information absent from the original Newton polytope; it does not replace lower-facet certification.

Allowing massive internal propagators would relax the standing massless multiaffinity assumption. The resulting quadratic edge-parameter dependence changes the possible cancellation jets and the local facet problem. It is a genuine extension of this framework, not another example in the present tour.

A Notation summary

Symbol	Meaning
δ	kinematic expansion parameter
$\boldsymbol{s}$	real kinematic coordinates; their physical domain $\mathcal{K}$ is specified by explicit inequalities
x_e	independent LP edge parameter, integrated over $(0,\infty)$
$\mathcal{C}_{\text{LP}} = \mathbb{R}^N_{>0}$	non-projective LP integration contour
$\mathcal{P} = \mathcal{U} + \mathcal{F}$	Lee–Pomeransky polynomial
$\deg_x \mathcal{U} = L,\ \deg_x \mathcal{F} = L+1$	massless graph-polynomial degrees; both polynomials are linear in each individual x_e
$\widehat{\mathcal{P}} = \mathcal{F} + x_0\mathcal{U}$	optional homogeneous $(N+1)$-parameter embedding
$m_i = c_i(\boldsymbol{s})\delta^{a_i}\boldsymbol{x}^{\boldsymbol{r}_i}$	monomial of $\mathcal{P}$
$\boldsymbol{r}_i$	length-N LP-parameter exponent vector of m_i
$\vec{r}_i = (\boldsymbol{r}_i; a_i)$	augmented exponent point
$\mathcal{F}_0$	strict fixed-$\boldsymbol{x}$ leading polynomial
$\mathcal{F}^{[\phi]}$	face selected by the preliminary alignment vector $\boldsymbol{\phi}$
$x_e = \delta^{\phi_e} y_e$	first transformation: asymptotic-order alignment
$\mathcal{F}_\star$	HRF starting polynomial, either $\mathcal{F}_0$ or $\mathcal{F}^{[\phi]}$
$\mathcal{S}_{\text{SL}}$	superleading monomial support selected by the scaling law
$\mathcal{F}_{\text{SL}}$	superleading cancellation sector
$\mathcal{F}_{\text{Obs}}$	obstruction polynomial
f_j	non-monomial cancellation factor
$\mathcal{I}$	ideal of independent cancellation factors at the common pinch
$\boldsymbol{\rho}^{(0)}$	provisional face-only HRF scaling in the aligned variables
$y_e = \delta^{\rho_e} z_e$	corrected second-stage HRF scaling after all LP layers are restored
$\boldsymbol{v}_{\text{HR}} = \boldsymbol{\phi} + \boldsymbol{\rho}$	composition of the two edge-parameter transformations
$\vec{v}_{\text{HR}} = (\boldsymbol{v}_{\text{HR}}; 1)$	certified total hidden-region vector; the last component is appended once and is not added stage by stage
W_{SL}	common individual weight of SL monomials
W_{HR}	first resolved nonvanishing LP weight after cancellation
$W_{\text{HR}} - W_{\text{SL}}$	cancellation depth
red terms	individually superleading sector at W_{SL}
blue terms	resolved physical leading layer at W_{HR}
black terms	all other layers, including intermediate ideal-valued cancellations

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